

PRATIK DESAI

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Summary

Full-stack web developer with a strong focus on **backend development**, specializing in **distributed systems and cloud computing**. Passionate about building scalable, **AI-enabled applications** that drive innovation and efficiency.

Education

Texas A&M University

Master of Computer Science (GPA: 4.0)

Aug. 2023 – May 2025

College Station, TX

College of Engineering Pune

Bachelor of Technology in Computer Science (CGPA: 8.56/10)

Aug. 2018 – May 2022

Pune, India

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, Ruby, C/C++, SQL, HTML/CSS

Frameworks: Spring Boot, Next.js, React, Node.js, Flask, Tailwind

Developer Tools: Git, Docker

Cloud Computing: AWS Services (EC2, S3, SQS, Lambda, DynamoDB), Azure, Google Cloud Platform, Big Data

Databases: MySQL, NoSQL, MongoDB, PostgreSQL, Redis, Firebase, Hadoop

Experience

Nuance Communications - A Microsoft Company

July 2022 – July 2023

Software Engineer

Pune, IN

- Architected and implemented a dynamic feature flagging system with **Spring Boot** and **PostgreSQL**, reducing deployment time by 60% and enabling real-time experimentation across multiple microservices.
- Enhanced API throughput by 20% by refactoring critical code paths, optimizing database queries, and integrating adaptive load balancing, resulting in a 35% reduction in server response times under peak loads.
- Refactored React components to adhere to MVC principles, increasing code reusability by 40% and reducing technical debt, which improved developer productivity and accelerated feature delivery by 25%.
- Boosted site performance by 40% by implementing React Query for intelligent caching and optimized data fetching strategies, reducing initial load times and enhancing user experience on high-latency networks.

Texas A&M University

June 2024 – Aug. 2024

Full Stack Software Developer

College Station, TX

- Led the end-to-end development of the OliviaKIDS educational platform using Next.js, React.js, TypeScript, Tailwind CSS, Zod, and Zustand, streamlining development processes and reducing build times by 20%.
- Engineered a scalable backend infrastructure with Prisma and PostgreSQL, handling 10x data growth efficiently and improving database query performance by 30% through advanced indexing and partitioning strategies.
- Streamlined deployment processes by integrating Docker, ensuring consistent and reliable environments across development, staging, and production, which reduced deployment failures by 50%.

Nuance Communications - A Microsoft Company

June 2021 – Aug. 2021

Software Engineer

Pune, IN

- Optimized cloud-based data storage using advanced indexing, partitioning, and compression techniques in PostgreSQL, achieving a 30% reduction in storage costs and 20% faster query performance.
- Led internationalization efforts, implementing support for 2 new languages, leveraging cloud-based localization services, and expanding application accessibility to global markets with a potential user base increase of 15%.

Leadership / Extracurricular

Teaching Assistant

Fall 2024 – Present

Texas A&M University

- Conduct lab sessions, providing hands-on experience with Ruby on Rails, Docker, and CI/CD practices
- Guide project teams through technical challenges and project milestones, offering support and expertise
- Evaluate and grade lab assignments, offering constructive feedback to enhance students' understanding and skills

Projects

StackSense | **Link** | *Flask, React.js, BERT, TensorFlow, LLM, Generative AI*

- Designed and implemented an optimized BERT-based model for tag prediction on StackOverflow, increasing F1-score from 0.548 to 0.567 through targeted fine-tuning and algorithmic enhancements.
- Implemented KNN collaborative filtering with Universal Sentence Encoder, accelerating and improving tag suggestions and question recommendations on StackOverflow.

WhatsNxt | **Github** | *React.js, SQL, RestfulAPI, Python, Flask*

- Engineered a multifunctional Chrome extension compatible with 5+ social media platforms, boosting user engagement through personalized features.
- Delivered features such as personalized auto-complete and auto-response, achieving an average model accuracy of 93%.
- Enhanced user productivity by engineering an auto-calendar link generation feature through improvements to an existing library.
- Led a cross-functional team to develop and implement the project, demonstrating effective communication and project management, ensuring alignment with objectives and timelines.

Employee Management System | *Spring Boot, RestfulAPIs, PostgreSQL, Spring Cloud, React.js, Node.js*

- Designed and developed core Microservices for Employee Management System, encompassing Employee, Payroll, and Attendance functionalities.
- Implemented a notification service using Twilio and RabbitMQ, enhancing communication efficiency and reliability within the system.

HTTP Server | **Github** | *Python, Multithreading, Concurrency*

- Built a high-performance HTTP server in Python, capable of handling 10,000+ concurrent connections through advanced multithreading techniques.
- Integrated logging and monitoring capabilities to track server performance and detect potential issues.
- Implemented GET, POST, PUT, and DELETE methods, achieving **low-latency response times** under test conditions.

Course Evaluation Summarizer | **Github** | *Ruby on Rails, ChatGPT API, PostgreSQL, React.js*

- Automated analysis of course evaluations through a web application, reducing manual processing time by 90%.
- Achieved key milestones including a user-friendly interface, Google OAuth login, and PostgreSQL database setup, laying the foundation for a robust application.
- Enhanced the system with AI-driven analysis using the ChatGPT API, improving data processing and stability for more accurate and efficient analysis.